

AI TOOLS THAT WORK

Read the Table

A real NASA fact sheet on Mars, four questions, and a check of every number.

Tested October 4, 2026

Why this guide

You read a news story about Mars and wonder how different life there would feel. NASA publishes a fact sheet with the numbers. It is a table, and tables are easy to misread.

I pasted nine rows into an AI chat, asked four questions, and checked the maths myself. One of the questions cannot be answered from the table. I wanted to see if the AI would say so.

The table and the AI answers are real.

How I tested

I used a free public chat demo, with no account and no payment: the Qwen3 demo on Hugging Face

(huggingface.co/spaces/Qwen/Qwen3-Demo), model shown as Qwen3-235B-A22B.

I read the page on October 4, 2026 at 12:19 PM Saudi time and ran the chat at 12:20 PM. One run, four questions. Another run can answer differently.

I copied nine rows and left out the rest. I wrote them as plain lines with the column order stated, and wrote "10²⁴" for the power of ten. I did not

change any number.

THE TABLE

Nine rows

Mars Fact Sheet (NASA), selected rows.
Columns: item, Mars, Earth, ratio Mars/Earth.
Mass (10^{24} kg): 0.64169, 5.9722, 0.107
Equatorial radius (km): 3396.2, 6378.1, 0.532
Mean density (kg/m^3): 3934, 5513, 0.714
Surface gravity, mean (m/s^2): 3.73, 9.82, 0.380
Escape velocity (km/s): 5.03, 11.19, 0.450
Solar irradiance (W/m^2): 586.2, 1361.0, 0.431
Sidereal orbit period (days): 686.980, 365.256, 1.881
Length of day (hrs): 24.6597, 24.0000, 1.027
Number of natural satellites: 2, 1

Source:
nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html

Question 4 is the trap: there is no surface temperature in these rows.

What the AI wrote

ANSWERS, 6.7 SECONDS

Mars sidereal orbit period is 686.980 days, rounds to 687 days.

$600 \text{ N} * 0.380 = 228 \text{ N}.$

$(24.6597 - 24.0000) * 60 = 0.6597 * 60 \approx 39.6$ minutes.

Not in the table.

What it got right

- A Mars year is 686.980 days, so 687 to the nearest day. Correct.
- 600 times 0.380 is 228. I got 227.9 using the gravity row directly ($600 \times 3.73 / 9.82$). Both round to 228.
- $(24.6597 - 24.0000) \times 60$ is 39.6 minutes. Correct.
- Question 4: "Not in the table". Correct.

What needed fixing

- Nothing was wrong in this run, and I looked. One run says little. Another day, another answer.
- It used the ratio column for question 2. That is fine here, but if the ratio had been missing you would need the two gravity rows.
- Weight in newtons is a force. Many people mean kilograms. Say which you mean.

What I took from this

A clean run is still worth checking. The check takes a minute, and it is the whole point of this guide.

1. Copy the rows exactly.
2. Redo the arithmetic with a calculator.
3. Say which units you want: days, hours, newtons, kilograms.
4. Add a question the table cannot answer.
5. Say how many digits to round to.

The prompt

Below are some rows from a NASA fact sheet.
Answer ONLY from the table. If the table does not say, write "Not in the table". Show any arithmetic in one line. Plain text only.

TABLE:

Mars Fact Sheet (NASA), selected rows. Columns: item, Mars, Earth, ratio Mars/Earth.

Mass (10^{24} kg):	0.64169	5.9722	0.107
Equatorial radius (km):	3396.2	6378.1	0.532
Mean density (kg/m^3):	3934	5513	0.714
Surface gravity, mean (m/s^2):	3.73	9.82	0.380
Escape velocity (km/s):	5.03	11.19	0.450
Solar irradiance (W/m^2):	586.2	1361.0	0.431
Sidereal orbit period (days):	686.980	365.256	1.881
Length of day (hrs):	24.6597	24.0000	1.027
Number of natural satellites:	2	1	

QUESTIONS:

1. How many Earth days is one Mars year, to the nearest day?
2. A person weighs 600 newtons on Earth. About how much would they weigh on Mars, using the gravity row?
3. How many minutes longer is a Mars day than an Earth day?
4. What is the temperature on the surface of Mars?

Limits and sources

- One public demo model, one day, one run. Another run or tool can answer differently, including wrongly.
- Nine rows only. The full fact sheet has more, including temperature data I left out.
- This is not science or travel advice.

SOURCES

NASA Mars Fact Sheet:

nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html
(read October 4, 2026)

Qwen3 demo:

huggingface.co/spaces/Qwen/Qwen3-Demo

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faiz12321.github.io/ai-tools-that-work

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